Comparison of Different Input Data for the Prediction of LED Solder Joints Using Artificial Neural Networks

Andreas Zippelius, Zubair Mohd, Maximilian Schmid, and Gordon Elger

***Abstract*—Scarcity of raw data is a major issue for applying data driven methods to reliability prediction, so making the best use of what is available is critical. This paper studies how different aspects of measurement data can be used best. Specifically, the reliability of the solder joint of LED packages is predicted based on Transient Thermal Analysis and Scanning Acoustic Microscopy data from a large measurement campaign. The impact of using full temporal information vs measurements at individual datapoints is investigated as well as the benefit of including the SAM data, and different ways of presenting the TTA information, either as a full curve or as expert-selected features. The impact of formatting categorical information of solder and LED package type as one-hot encoding or using embeddings is considered. Finally, the performance for Pass/Fail predictions of the best identified model with a model architecture developed on a similar dataset is compared. We identified the most relevant sources of information for predicting the behavior and the best format for the data, which helps guide the choice for future model architectures.**

***Index Terms*—Reliability, LED, artificial neural network, convolutional neural network, transient thermal analysis, scanning acoustic microscopy.**

I. Introduction

RELIABILITY is one of the major requirements for microelectronic systems in any application, especially for safety-relevant electronic components, power and opto-electronics. Reliability and lifetime analysis seek to build models that quantify the risk of failure for a system at a given moment in time based on the available information to predict the remaining useful lifetime (RUL). For microelectronic systems and especially high power LED packages, the lifetime is strongly influenced by the operating temperature. A common source of damage is thermomechanical stress due to differing expansion and contraction during temperature changes of different parts, expressed in the mismatch of the coefficients of thermal expansion (CTE). This damages the structural integrity, typically focused on the solder joint, through repeated plastic deformation resulting in material fatigue. Increases in the operating temperature will also negatively impact the performance characteristics of LED package such as efficiency and color. This makes the matter particularly critical for LEDs, since there is the negative feedback loop. A damaged joint has worse heat transfer, producing a higher operating temperature and therefore worse efficiency, which further raises temperature, which together also increase the stress on the joint and speed up the damage. Managing the thermal properties of high performance LED packages is very important to archive that high performance.

The reliability of Tin-Silver-Copper (Sn-Ag-Cu, SAC) solders is a topic of great research, as these represent the current state-of-the-art for non-lead solders [1], [2], [3]. These also include a great deal of effort understanding the effect and evolution of the microstructure [4], or the different usage conditions [5]. In this paper, a previously collected dataset of aged LEDs is used [6], [7].

Transient Thermal Analysis (TTA) is a method based on the heat transfer through the component and a fast and non-destructive method to detect cracks and delamination in microelectronic systems [8], [9], [10]. The increase of the thermal resistance is also a useful failure criterion for solder interconnects [11]. A useful supplement is Scanning Acoustic Microscopy (SAM) [9], which non-destructively produces a 3D or 2D image that can be easily interpreted by humans to estimate the state of the joint, adding spatial information lacking in TTA measurements, and unlike standard X-Ray can detect solder cracks.

In this paper, the aging of LED solder joints is predicted and the use of different types of input data for the prediction is compared. TTA is well suited for lifetime prognostics [12], [13] and is used as the main basis for the prediction, either in the form of a processed feature vector or as a full curve. In this paper, they are supplemented with SAM measurements. The aging is modeled using neural networks, which have been used for this and similar tasks previously [14], [15]. Two prediction approaches are compared: predictions based on the data taken at one cycle, and predictions based on all measurements up to that point. This allows a comparison of the impact of using the aging history for the prediction. Further, the impact of different data inputs is considered, either in the types or the preprocessing. These investigations are done for a regression task, that is prediction

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Andreas Zippelius, Maximilian Schmid, and Gordon Elger are with the Institute of Innovative Mobility, Technische Hochschule Ingolstadt, 85049 Ingolstadt, Germany (e-mail: Andreas.Zippelius@thi.de).

Zubair Mohd was with Technische Hochschule Ingolstadt, 85049 Ingolstadt, Germany. He is now with Volkswagen, Kassel, Germany.

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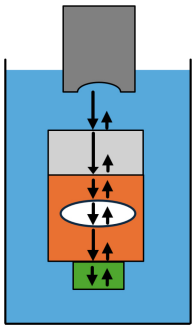


Fig. 1. Cracking of an example solder joint over the cycles, with cracks or voids showing in white. As typical, it proceeds from the corners inwards. Note that a black area can still be damaged if the crack runs at an angle and so the reflection does not reach the transducer again.

of a continuous variable. Both the possible performance gain of adding SAM images is considered as well as the how much accuracy is lost when restricting the model to just expert selected features instead of the whole TTA data curve. Finally, the results of the best model is compared to the results obtained by another model developed on a dataset of very different LED package (Wafer Level LED), in terms of categorical pass/fail prediction. Categorical prediction are used to reduce the influence of differences in dataset.

The next section will explain the data basis (measurement methods, study setup, measured components). Section III will describe the background for the machine learning approach and the architecture of the models. Section IV discusses the results.

II. Data Collection

In the following section the data collection process is briefly described. For a more in-depth description of the components, soldering, and initial characterization see [10]. First, the measurement techniques used to collect the data in this paper are discussed, and then the components and study design is described.

A. Scanning Acoustic Microscopy

Scanning Acoustic Microscopy (SAM) is a non-destructive measurement method based on the same principles as the ultrasound used in medicine that allows inspection of the internal structure of components using focused sound waves. Parts of the emitted acoustic wave are reflected at any change in acoustic impedance, such as an interface between two different materials. This is shown in Fig. 1. Based on the timing and strength of the reflections, the composition of the specimen can be reconstructed for a single position. The complete image is produced by scanning across in a grid of the desired x/y-resolution [9], [11]. The specimen is placed in water to help transmission from emitter to specimen and back, thus after measurement the part has to be dried. This plus the time required for the scanning means that a SAM image requires a significant amount of time and work investment, and in this study was only taken at four different timesteps.

SAM can show voids, non-wetted solder areas and cracks, since the interface to the material (water, or gas and resin from the flux) filling the crack/void produces a reflection. The progression for one sample LEDs over the aging cycles is shown in Fig. 2. However, resolution is worse than X-Ray, and depends on the acoustic frequency. This is a trade-off, since higher frequency allows better resolution but lower penetration. For the components used here, a higher frequency with better resolution was not possible due to insufficient penetration depth.

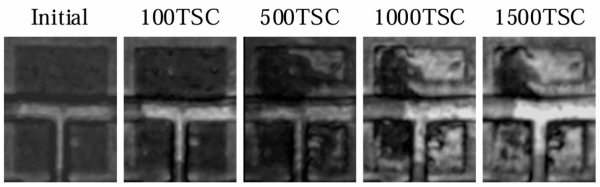


Fig. 2. The working principle of the SAM measurement. Sound waves, shown as arrows, are sent out from the transducer at the top, and get reflected at each material layer (not drawn to scale here), or on material defects. Complex structures, such as most electronic components including LEDs, scatter so much of the sound waves that they cannot penetrate through such layers. Here, the component is analyzed from the back, passing through the IMS to reach the solder joint.

In this paper, 2D images at the depth of the solder joint (C-mode) are used, though SAM is in principle capable of full 3D images. However, the z-resolution is not high enough to justify the large increase in computational resources needed to use the full 3D data, as different parts of the solder joints do not show meaningful differences with our resolution. The frequency of the sound waves used was 50 MHz. Higher frequencies offer better resolution but lower penetration. 50 MHz is the highest frequency that can penetrate far enough, since the sound has to travel through the PCB. Going through the LED is not possible, since the internal structure scatters too much to let the waves reach the solder joint.

B. Transient Thermal Analysis

A quicker and easier way to characterize solder joints than SAM, though lacking spatial resolution, is utilizing by the heat flow through the joint via Transient Thermal Analysis (TTA). The measurement is also non-destructive, can be done in a few second, and only requires a precise voltage measurement. Analysis of the raw data is also more straightforward, though still requires some processing. LEDs in particular are well suited to this measurement method, heat produced from usage can serve as the heat source required for the measurement. This method has been used for measuring joint integrity during accelerated aging in other works [6], [16].

TTA measures the time-dependent thermal impedance Z_{th}, i.e., how easy heat can flow from the source to the sink as a function of time. Defects such as cracks in the solder impede the flow by reducing the contact area. The further a defect is from the heat source, the longer the heat flow takes to reach it, and the later it shows up in the signal, which allows roughly locating the defect on a position on the thermal path [12], [13].

The standard for TTA measurements is described in JEDEC Standard JESD51-14 [17] and the used TTA tester is described in [18].

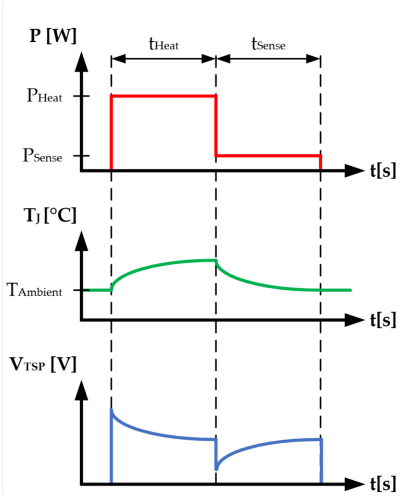


Fig. 3. The heating and measurement phases during a TTA measurement. Shown are the electric power used, device temperature, and applied voltage.

To perform the measurement, the temperature over time $T_J(t)$ at pn-junction is determined via the voltage $V_f(t)$ at a fixed current using the temperature sensitivity SEN

$$T_J(t) = \frac{V_f(t)}{SEN}. \tag{1}$$

$Z_{th}(t)$ can then be calculated from the change in temperature since the start of the measurement $\Delta T_J(t) = T_J(t) - T_J(t_0)$ using the change in heating power ΔP_{th}

$$Z_{th}(t) = \frac{\Delta T_J(t)}{\Delta P_{th}}. \tag{2}$$

In our measurement setup, a TTA measurement consists of a heating phase and a measurement phase, visualized in Fig. 3. Only the cooling phase is considered due to higher signal quality. For better accuracy, this is repeated measurement is repeated 10 times and averaged.

The thermal resistance R_{th} is the steady state value of the thermal impedance

$$R_{th} = Z_{th}(t \to \infty). \tag{3}$$

A useful way to further analyze the data is to consider the derivative with respect to the logarithmic time $z = \log(t)$. Cooling can be, roughly, approximated as an exponential decay, since the heat flow is proportional to the temperature difference to the environment, using a logarithmic timescale makes the dynamics more clear.

The logarithmic derivative is defined as

$$b(z) = \frac{d}{dz} Z_{th}(z) = \frac{d}{dz} \frac{\Delta V_f(z)}{SEN * \Delta P_{th}} \tag{4}$$

Using $B = \log(b)$ makes interpretation easier and can be interpreted in terms of R_{th} and will be used primarily. One advantage is that it B is less susceptible to noise/variation of the measurement apparatus. Both changes in temperature sensitivity of the measured joint (SEN) and changes in thermal path after the joint (most often due to the thermal interface material between the PCB and the heat sink) can be more easily compensated for. For details see [13].

TABLE I
COMPOSITIONS OF THE USED SOLDERS

Solder	Additive material					
	Ag	Cu	Sb	Bi	Ni	In
SAC105	1.0	0.5	-	-	-	-
SAC107+BiNi	1.0	0.7	-	1.6	-	0.2
SAC305	3.0	0.5	-	-	-	-
SAC327+Sb	3.2	0.7	5.5	-	-	-
SAC387+SbBiNi	3.8	0.7	1.5	3.0	0.15	-

In practice, further computation is needed to deal with various sources of noise. To compute Z_{th} the data is first smoothed with a running mean filter (window length 15) and the early time range of 0 to 20 μs is extrapolated using a $\sqrt{t}$-fit in the region of 20 μs to 40 μs to deal with the interference of switching from heating to measuring currents [10].

To compute the derivative b, Z_{th} is interpolated with cubic splines to 256 grid points uniformly distributed in z-space in the range between ln(2 μs) and ln(1.95 s) and then filtered with a Tikhonov regularization filter [19] with $\lambda = 0.00001$, since the differentiation produces high frequency noise.

For further use with the network, we use three hand-selected features: B_{max}, the maximum value of $B(z)$, z_{max} the position in z of the peak B_{max} with $B(z_{max}) = B_{max}$, and the translation factor between V_f and b in equation (4), $unit = SEN * \Delta P_{th}$. $unit$ is measured in $\frac{W*V}{K}$. z and B, and therefore z_{max} and B_{max} are logarithmic quantities and therefore unitless, but they can be considered in terms of s and $\frac{K}{W*s}$. B_{max} is also used to define the failure. Previous work has shown that these features capture a significant part of the available information [14].

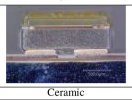
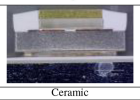
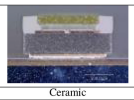
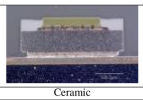
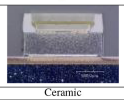
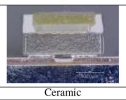
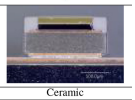
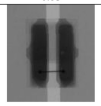
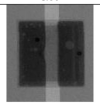
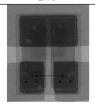
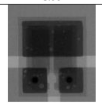
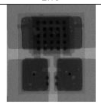
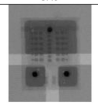
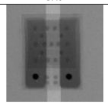
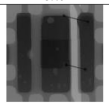
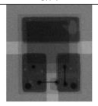
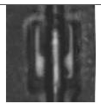
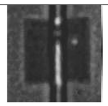
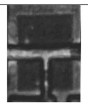
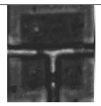
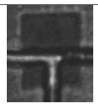
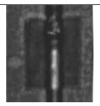
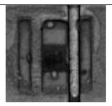
C. Study Setup

For the base data, nine different LED packages were soldered with five different solders (SAC105, SAC+BiIn, SAC305, SAC+Sb, SAC+SbBiNi, see Table I) to Aluminum insulated metal substrate printed circuit and aged with thermal shock cycles. Each solder/LED type combination contributes 40 LEDs, for a total of 1800 LEDs. The study is described in full detail in [6], [7]

The packages use different LED technologies. Two packages used lead frame submounts, while the other seven use ceramics. A listing of the LED properties is shown in Table II, taken in a modified version from the initial paper describing the study in detail. Reference [6] also contains a more detailed description of the structure of the LEDs, the data collection methodology, the and study as a whole, which for brevity we do not discuss in depth here. FC-SP1 and VTF1 have copper carriers, which have a much smaller CTE mismatch with the metal substrate and therefore less thermal stress, resulting in a significantly longer lifetime. FC-GB2 and FC-GB3 have a very thin copper ceramic metalization, which in the study proved to significantly reduce the lifetime. The failure modes observed were solder cracking, and a few LEDs showed damage to the phosphor layer which can be distinguished from cracking based on the TTA. For a more detailed analysis see [6] and [7].

The LEDs were subjected to shock cycles of -40°C to 125°C with dwell times of 30 minutes at the high and low temperatures in an air-to-air chamber according to JEDEC standard

TABLE II
LEDs Used in the Study With Most Critical Properties

Name	FC-SP1	FC-SP2	FC-SP3	FC-SP4	FC-GB1	FC-GB2	FC-GB3	VTF1	VTF2
Classification	Automotive	General	Automotive	Automotive	Automotive	Automotive	Automotive	Automotive	Automotive
Image									
Cross-Section									
Carrier Material	Copper	Ceramic	Ceramic	Ceramic	Ceramic	Ceramic	Ceramic	Copper	Ceramic
Pad Count	2	2	3	3	3	3	2	3	3
Size [mm]	1.92	2.56	5.0	3.03	4.37	2.61	2.61	14.06	2.85
Pad-Size [mm^2]	0.81	1.80	2.79	1.86	2.75	1.45	1.49	5.40	1.74
X-ray (not scaled)									
SAM (not scaled)									

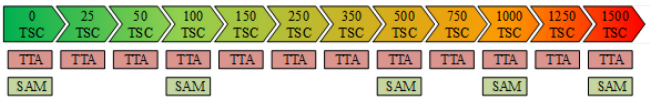


Fig. 4. Measurement intervals during aging.

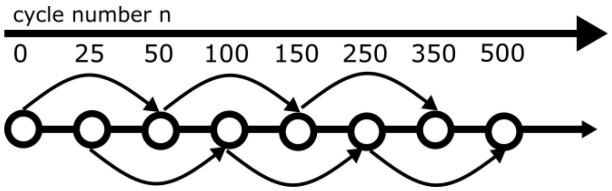


Fig. 5. Sketch of prediction. The circles are the states for different cycles. Each arrow is one prediction based on the available information (current plus past states) for a certain number of steps (two here) into the future. Since the number of TSC between measurements grows towards later cycles, later predictions span a greater number of cycles.

JESD22-A104D Profil G [20]. Measurements were taken at the cycle counts shown in Fig. 4. Each TTA measurement also includes a temperature sensitivity measurement to determine *SEN*.

The dataset is reduced by 90 LEDs for 500 cycles and 180 for 1000 cycles as two LEDs per Solder/Component combination were used for cross sections. These samples were discarded for this study.

III. Artificial Neural Networks; Background and Implementation

The use of machine learning allows for predictions based on the individual behaviors of the LEDs rather than the ensemble, and therefore greater potential accuracy, especially in regard to outliers or multifaceted populations.

ANN have been successfully used for lifetime prediction in general and LEDs in particular. Liu et al. [21] used an ANN for more accurate predictions of single temperature lifetime for multi-chip high-power LEDs. Chao et al. [22] predicted the development of the lumniance of LEDs using artificial neural networks. Fan et al. [23] developed a method to predict the full-spectrum white LED under different operation modes which incorporates an ANN. Machine learning approaches also proved to be more accurate than classical methods such as Weibull to predict the reliability of solder joints in the study by Yi and Jones [24]. LSTM architecture was used for a condition monitoring based failure prediction by Achouch et al. [25] for compressors. Ibrahim et al. [26] used LSTM based architecture to predict the resistance during power cycling.

Here, we focus on regression on a time series and compare different types of input data. To better contextualize the performance in relation to the difficulty of the task, we also perform a Pass/Fail prediction using the best model found here and the model developed and tested for a prior LED dataset described in [14].

A network architecture of Long Short-Term Memory (LSTM) layers and standard dense layers is used for the main computation. A dense layer combines the output of the previous layers in a manner independent of the preceding inputs for the final prediction and can be considered a matrix multiplication followed by a nonlinear activation function. The LSTM layers are an extension of the dense layers which have an additional memory for previous inputs, and as part of the training, the network also learns to manipulate this memory.

By using an LSTM based architecture, a prediction for an LED is made every time more measurement data becomes available. This can be for the next measured state, or the state the state after, or two after, etc. A sketch matching of step to cycle when predicting one step in advance is shown in Fig. 5. Note that the length of a step is not constant; later steps cover a larger number of TSC.

The linear rectifier is used as the activation function σ,

$$\sigma(x) = \max(0, x) \tag{5}$$

To extract useful features from 1D or 2D data, the standard approach of convolutional neural network (CNN) and pooling layer is used. As the name says, a convolutional layer performs a convolution of the input with a kernel, which contains the trainable parameters. This can be done with either 1- or 2-dimensional kernels, depending on the dimension of the input. Typically, one convolutional layer contains many different

parallel kernels trained to detect different features in the image, which get combined into one output image (similar to how an RGB color image consists of three images that each show a single color). This it typically followed by a Pooling layer, which condenses the information of multiple pixels and reduces image size. Deeper convolutional layer detect increasingly larger and more complex structures. Here, we use Max Pooling, with takes the max value in box as the output. A Max Pooling Layer with stride of 2 will therefore produce with half the dimension sizes, and a quarter of the number of pixels.

The advantage of CNN layers is that they automatically consider spatial relationships in the data by nature of how a convolution works (only the nearby points covered by the kernel are used), and that they have orders of magnitude fewer training parameters (because only local points are considered, instead of all as with the dense layer). The lower number of parameters makes training faster and more robust. Since the output of a CNN is another image, the computational process is also more easily understood, which is difficult for typical NN models.

We use three different types of input:

- The feature vector, mainly extracted from the TTA curve based on physical understanding of the system.
- The full curve of $B_n(z)$, which consists of 256 values grid points
- The SAM image closest to the current cycle, since SAM was not measured every cycle

The main feature vector for cycle n consists of the following values

$$\left(\Delta z_{max,n},\, \Delta B_{max,n},\, \mathrm{unit}_n\right).$$

$\Delta z_{max,n}$ and $\Delta B_{max,n}$ are the change in the peak position and height compared to the 0-cylce measurement at cycle n, and unit_n is the value to translate from the measured voltage to the impedance $Z_{th,n}$, a combination of the temperature sensitivity SEN and ΔP_{th}.

These values are rescaled to a mean of 0 and a variance of 1 to aid training, since it makes the different parameters more comparable, and network doesn't have to learn the comparative sizes or rescale to the "expected size". For example, a very large negative value could produce problems with the activation function, since it would require much training to see an effect.

When using the SAM images as input, the feature vector is extended with a value that indicates how long ago the current SAM image was taken. This is given as 0 for a current image, and adding 0.1 for each measurement it is in the past. The value of 0.1 is chosen for scaling compared to the other values.

Finally, we also consider the effect of categorical information, specifically the solder, the component or both. Two possible ways of encoding this information are considered. Option One is as a one-hot encoding per category, which for a category with n options means a vector of n-1 zeroes and a single one, where the position of the 1 indicates which category it falls into. This method is simple and requires no separate training, but does not scale well to large categories. The input data is also quite sparse. Option Two is to utilize embeddings, which first transform the category into a vector (here 4D) which is then given to the rest of the network. This method requires training to create/learn these embeddings, but can pack more information into the input. Different directions carry meaning and can be added, and this method scales better to larger categories since only a logarithmic increase in embedding vector size is required.

The output of the network in the regression task is ΔB_{max} of the target cycle. The target cycle is the current measurement step n plus the prediction length, see Fig. 5. We consider predictions up to 6 measurement steps into the future.

For the categorical Pass/Fail prediction, a component is considered failed when ΔB_{max} at the target cycle is greater 0.3. The exact threshold value mostly only shifts the "official" death forward or backward, so long as it is within reason. The influence of the choice of failure condition was considered in a prior work here [14]. We use a binary categorical prediction. The network produces a value between 0 and 1, where 0 means a total certainty the component has not failed, 1 a total certainty that the component has failed, while values in-between a greater or lesser degree of certainty towards either possibility. The use of a continuous output value makes training much simpler, since computing the gradient of a non-continuous value is a great deal more complicated.

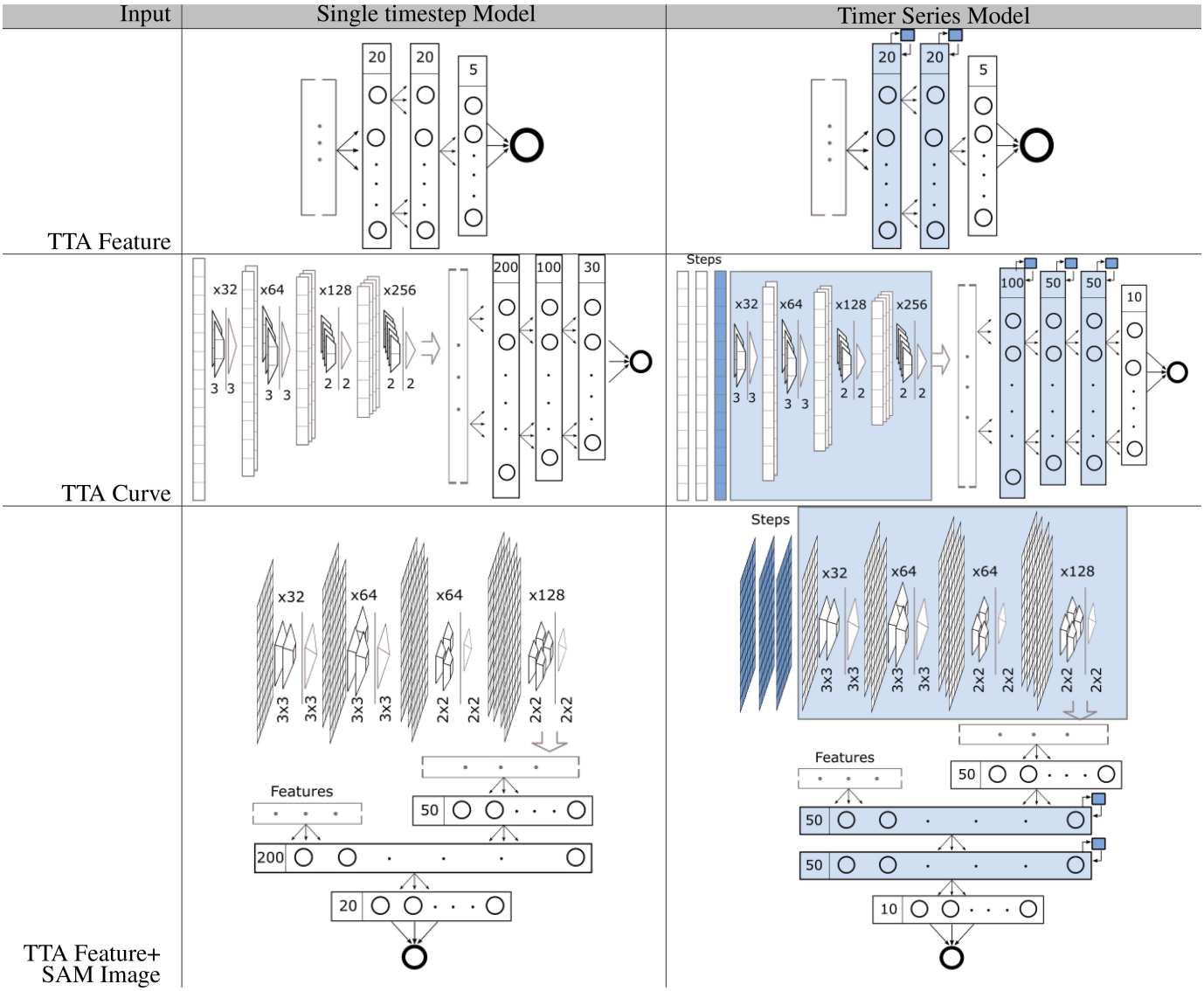
The models used in the paper are shown in Table III. First column is the architecture for predictions based on a single data point, second column is the model using a time series. The first row models use only the feature vector, the second row uses the full TTA curve, and the third row uses the feature vector supplemented with SAM images. The blue parts take the stepwise inputs from the time series. The box in fully-connected and LSTM layers shows the number of units in that layer. For the CNN layers, the number to the right shows the number of kernels (e.g., x32), which is followed by the size of the kernel and the size of the max-pooling layer. These are given as a single number for the TTA curve, or a two numbers of the 2D SAM images. Rectified Linear Units (ReLU) [27] are used as the activation function for the dense and LSTM layers except for the final output layer.

For the categorical task, we use two models, both making use of the time series information. One was a reference point taken from the prior work on a separate dataset [14]. It consists of three LSTM layers of thirty neurons each with ReLU activation, a dense layer of thirty neurons with ReLU activation, and a final dense layer to produce the output with a sigmoid activation function. The other model is the Feature and Image model in the table, see Section III, as the improved version. We use categorical information for both solder and LED. We use one hot encoding for both models to make them more comparable to the prior work in [14], since embeddings would be a significant change to the architecture.

The network is trained with the ADAM optimizer [28] using a learning rate of 0.005 and a batch size of 16. The mean square error is used as the cost function for the regression, while the binary cross entropy without label smoothing is used for the categorical prediction. The training parameters were initialized with the default settings for Tensorflow. This means the biases for all layers were initialized with

TABLE III

TABLE OF MODELS USED FOR REGRESSION. THE ROWS SHOW DIFFERENT INPUT DATA, FIRST JUST A VECTOR OF LED FEATURES, SECOND THE FULL TTA CURVE, AND FINALLY A COMBINATION OF SAM IMAGES AND TTA FEATURES. THE FIRST COLUMN SHOWS MODELS THAT TREAT THE MEASUREMENT OF EACH TIMESTEP INDEPENDENTLY, THE SECOND COLUMN SHOWS MODELS THAT TAKE THE INPUT AS A TIMESERIES. THE PARTS WITH MEMORY ARE MARKED IN BLUE. THE CONVOLUTIONAL PART (EITHER 1D FOR TTA OR 2D FOR IMAGES) TREAT EACH IMAGE INDEPENDENTLY, BUT THE RESULTS ARE USED IN THE CONTEXT OF THE TIMESERIES. THIS IS INDICATED BY THE LARGE BLUE BOX



zeroes, the weights were initialized from a uniform Glorot distribution [29], and the memory weights with an orthogonal initializer [30].

As is common procedure, the dataset is randomly split into training, testing and cross validation sets in ratios of 75%\10%\15%. This split ratio is enforced for each combination of solder and component individually. Theses behave quite differently and the sample set is too small to rely on the law of large number to ensure adequately comparable sets without enforcing an equal distribution. The training of the network is done on the training set and the final performance evaluated on the test set. All results shown are from the test set. The cross-validation set is used to control the training length and check for overfitting. Training is stopped once the model shows no improvement over 8 epochs for the models using CNN layers, or 20 epochs for those without (top row of Table III). The average number of training epochs before stopping is around 200 for the models using only TTA features, and around 40 for the models using a CNN architecture.

IV. RESULTS

In this section we discuss the different results and consider the influence of various design decisions.

We use three metrics to measure the performance of the model. These measure how good the predictions for the individual LEDs are in aggregate. For regression, we use the coefficient of determination R^2 and the Mean Absolute Error (MAE). R^2 is defined as

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_i^N \left(y_i - y_{p,i}\right)^2}{\sum_i^N (y_i - \bar{y})^2} \tag{6}$$

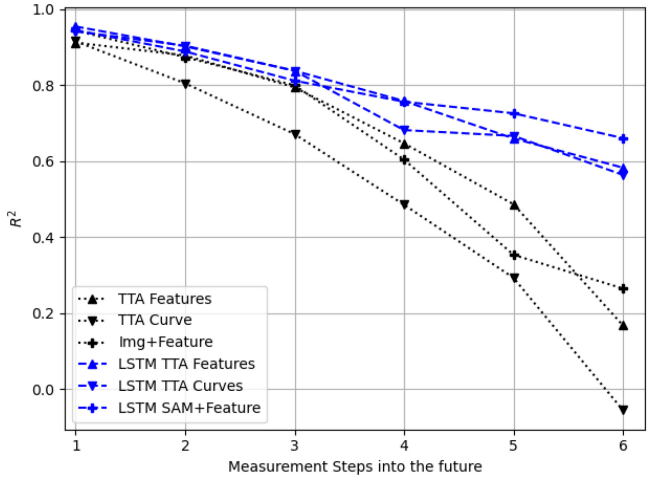


Fig. 6. Single prediction vs time series. Shaded blue is time series and corresponds to the right column in the model Table III, dotted black is single prediction and corresponds to the left column in the model table. The different inputs are shown with different markers and correspond to the rows in the model table. Each datapoint is the average performance of 4 trained models.

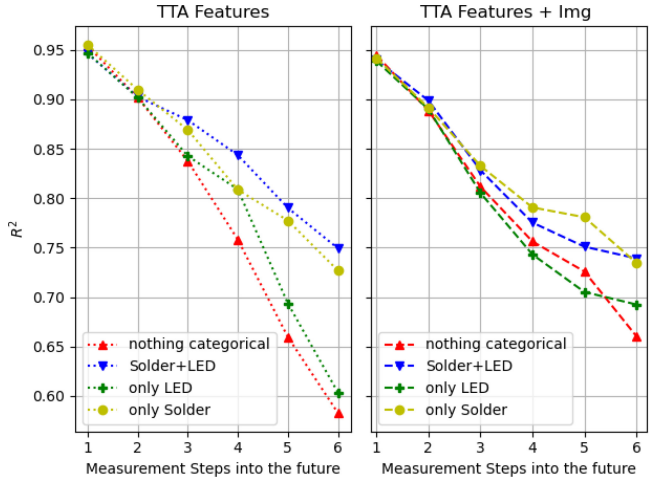


Fig. 7. Usage of categorical information. Left plot if for TTA features, right plot for TTA features supplemented with SAM images. Compared are models which have in addition also the solder, LED or both given as a one-hot encoding. All models here are using time series information.

with the SS_{res} the Sum of Squares of the residuals, i.e., sum of squared error, and SS_{tot} the total Sum of Squares, which is the variance except for the normalizing $1/N$ term. y_i is the true value for sample i, $y_{p,i}$ is the predicted value for sample i, and $\bar{y}$ is the mean of y. The model is perfect for an R^2 of 1, and as good as the mean of the dataset for an R^2 of 0. An R^2 score of 0.9 means that the model explains 90% of the variation in the dataset. The Mean Absolute Error is defined as

$$\text{MAE} = \frac{1}{N}\sum_{i}^{N} |y_i - y_{p,i}| \tag{7}$$

and describes the average distance of the prediction from the true value.

A. Effect of Time Series Prediction

We compare the effect of using only a single data point vs using the full times series. The result is shown in Fig. 6. It is clear that the time series information is quite important, especially for long range prediction. The performance is strictly greater for all models and all time distances.

The difference is fairly small for small distances but grows to very large sizes for long spans. We can conclude that the times series information is very important to estimate the more complex long-range behavior. One surprise is the poor performance of the single TTA Curves, since these perform significantly worse than everything else, even though more information is contained therein, and the main information of the feature vector is contained in the curve. This is likely due to the extra model complexity making training difficult given the size of the current dataset. The model runs into overtraining issues if trained for longer.

We can further see, that for single step prediction, the addition of the SAM features does not improve performance. This is contrary to the time series prediction, where SAM images provide a small improvement for very long-range predictions. Still, the fact that only the TTA features, even lacking temporal context, allow for a solid short-range prediction shows that a majority of the short-term behavior is dominated by fairly simple dynamics.

B. Categorical Inputs

The effect of the categorical information is shown in Fig. 7. We focus on the case of time series prediction, since this delivers the better performance as discussed in the previous section. We used embedding for this model as it performs slightly better, which is further discussed in the next Section IV-C. In case of solely the TTA features, we can see only small differences for the short-range behavior. For longer spans, the models separate into two major groups: The inclusion of solder information, or a lack. This is because the solder is very difficult to compute from just TTA measurements but does impact the aging strongly beyond just a crack percentage. For example, the difference in fatigue resistance between different alloys.

The addition of LED information provides a small but notable improvement to prediction. This is also expected, because the component type has a significant impact on aging due to the different stress produced in the pad and different pad shapes, but can only somewhat be deduced (because initial values are consistent across components).

Adding the SAM data in the left plot shows the components much closer together. Adding only LED information shows very little improvement in this case, because this information is already contained in the pad shape. However, solder information is still a valuable addition.

Of interest is also how the error is distributed for the different target cycles. This is shown in Fig. 8.

We can see that the error is smaller if the stepsize is smaller, which is expected since the task itself is different. For larger steps the error also grows quicker. The growth of the error is mostly the same for both models, except for the final cycles, where the model with SAM perform slightly better. The SAM offers less improvement for larger prediction length, perhaps

for 1 and 6 steps ahead, for two different models.

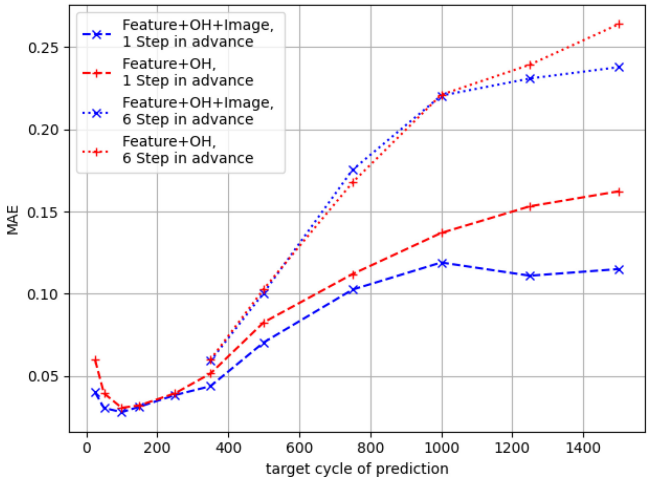


Fig. 8. Mean Average Error (MAE) for different target cycles. The datapoint is shown for the cycle the prediction is made, not the cycle of the information that is used for the prediction. Dashed lines are for six steps in advance, dotted lines are for the one step in advance. No prediction.

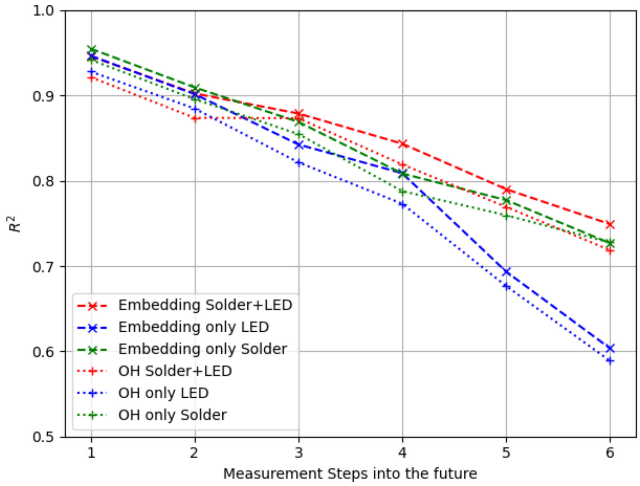


Fig. 9. Difference between the two studied ways to encode categorical information. Color describes which information is conveyed, the dashed lines with xs are using Embeddings, the dotted lines with pluses are using one-hot.

because the SAM image, which is not taken for all cycles, becomes too outdated. One interesting facet is the error starts higher and dips down. This is both because the model first has to collect information on the behavior of the individual LED, and from burn-in process that mean that the initial behavior of the LEDs is quite variable and more random.

C. Formatting Categorical Input

Two possibilities of how to convey the categorical information to the network were considered, either as a one-hot encoding or using embeddings. The results are shown in Fig. 9.

It's clear that embeddings perform better than One Hot encoding, though the difference is rather slight. For sufficiently small datasets, the added complexity and training required to using embeddings is not worthwhile. However, this dataset turns out to be larger than this threshold. This was something

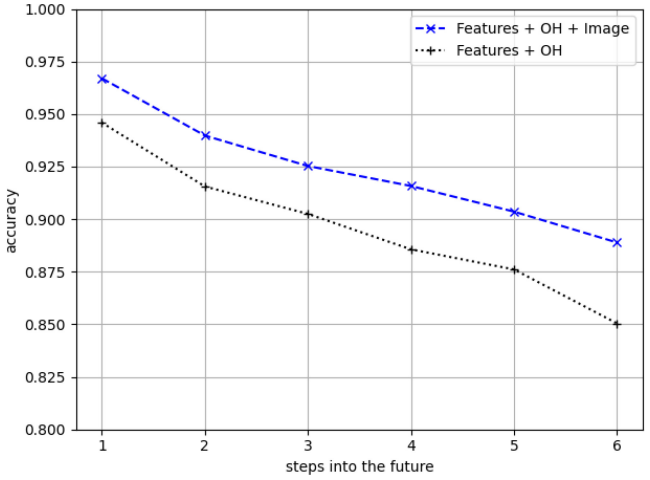


Fig. 10. Comparison of the prior model and best from this paper in the categorical prediction. Dashed blue is the new model integrating SAM information, while the dotted black line is the purely LSTM architecture used in the last paper. The accuracy is the proportion of correct prediction.

of a surprise, and initial testing was actually performed with one hot encoding before embeddings were checked for comparison.

One further advantage is in fact a better training behavior. There were some issues with the prediction in the initial timesteps ($0 \rightarrow 25$, $25 \rightarrow 50$) performing very badly. The usage of the embeddings actually improved training in this regard, though performance here is still notably worse than later on, since the available information is limited and the behavior of the LEDs rather erratic.

D. Pass/Fail Prediction

Often, what we care about in reliability is not so much the exact value of some output, but simply whether the component is still in an acceptable state. Quality assurance at End-of-Line is one use case. Condition monitoring, e.g., for safety critical aspects, produces multiple measurements over time and can benefit from the increased accuracy of the LSTM-based models. With the fail/pass prediction it can be guaranteed that within the next time interval the device will not fail. For a use case the key performance indicator would be measured, for example annually, and it would ensure that no failure occurs during the time for the next inspection. Finally, reduction of testing times by stopping aging tests early can help speed up the design loop.

A categorical prediction of pass or fail is therefore also of interest. Further, performance of an architecture is relative to the dataset, since not all datasets (and the underlying behavior) are equally complex. Comparison is desirable, but not trivial. A pass/fail prediction can simplify at least the aspect of the output to be directly comparable.

Here we compare the best model identified above with a model developed previously. We cannot train the model developed here on the older dataset, since it lacks SAM data, so we train the architecture used previously [14] on the current dataset. The results are shown in Fig. 10.

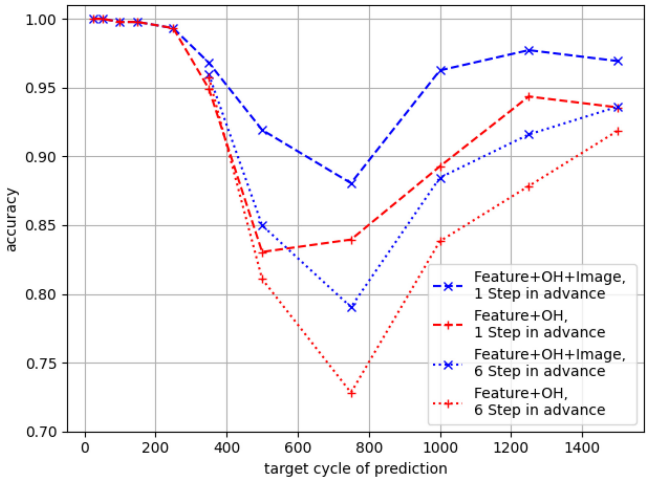


Fig. 11. Accuracy for different target cycles. The datapoint is shown for the cycle the prediction is made, not the cycle of the information that is used for the prediction. Cross are predictions 1 measurement step in advance, plusses are predictions 6 measurement steps in advance. Dashed lines are for the model using image data developed here, dotted lines are for the model using the previous architecture.

We see a high overall accuracy, even for very long term predictions. Though this needs to be relativized by the fact that the initial and final steps are fairly easy, since the LEDs will always start good, and past a certain point will clearly have failed. This can be seen in Fig. 11, which shows the accuracy for different target cycles.

We can see both the very high accuracy in the initial phase where not much has happened, and the also relatively high accuracy in the final phase where most components have failed. In the middle around the typical failure time is the region of highest uncertainty. Even so, the prediction can be made with an accuracy of around 80%, using just information from the first 200 cycles. We can see that the usage of SAM images shows a clear improvement.

V. Conclusion

We considered the influence of various design decisions on the accuracy of a artificial neural network model for the prediction of the state of LED solder joints.

By far the biggest impact is the inclusion of time series information using the LSTM. Without it, long term predictions are very weak, $R^2 < 0.3$ in the best case corresponding to a weak case and no success with in the worst case. With it, a somewhat useable prediction is possible though the accuracy is limited by the fact that 6 prediction steps corresponds to more than 1000TSC for the final predictions. The effect grows smaller for shorter timescales, where the immediate dynamics dominate. However for long time prediction, the information from previous times steps is critical for the performance of the network, and should be made available for a prediction model.

We found that a small set of TTA features already includes the majority of information about the state of the LED, especially with the extra context of the solder and LED type. A full curve does not improve performance. SAM images do, but the improvement is only notable for very long time scales. Part of this improvement is identifying the LED type when this information is not provided, though for these a substantial information R^2 gain of ≈ 0.1 is possible. This shows that having the information available in some shape is most important, but there is still an effect on exactly how it is available.

Adding information that cannot be easily deduced from the TTA features to the prediction provides a boost in performance as well. This includes solder type and LED type in our case, but could include other factors, e.g., PCB type, for other applications. With the right design decisions for the network, a reasonable prediction (R^2 of 0.75) can be made for 1500 cycles based on the first 250, which has the potential of strongly reducing the testing time required during aging.

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